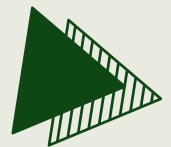
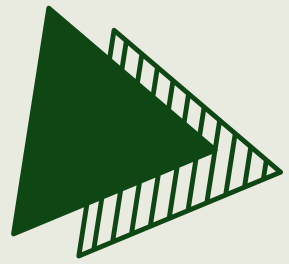


CROSS-PLATFORM SOCIAL MEDIA ENGAGEMENT ANALYSIS

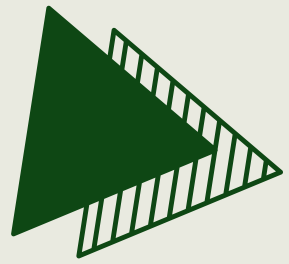


Fanshiqi Zhao



Data Resources

Platform	Source Type	Source URL	Sample Size	Key Variables	How Used in Analysis
TikTok	Kaggle JSON Dataset	https://www.kaggle.com/datasets/erikvdven/tiktok-trending-december-2020/data	1000+ videos	views, likes, comments, shares, hashtags	Viral short-form engagement
Instagram	Kaggle CSV Dataset	https://www.kaggle.com/datasets/kundanbedmutaha/instagram-analytics-dataset	30,000+ posts	likes, comments, reach, impressions	Image/social engagement
YouTube	YouTube Data API	https://console.cloud.google.com/apis/credentials?hl=zh-cn&project=youtube-data-project-489303	300+ videos	views, likes, comments, publish date	Long-form video engagement



Overall Engagement Rate Comparison

KEY FINDING

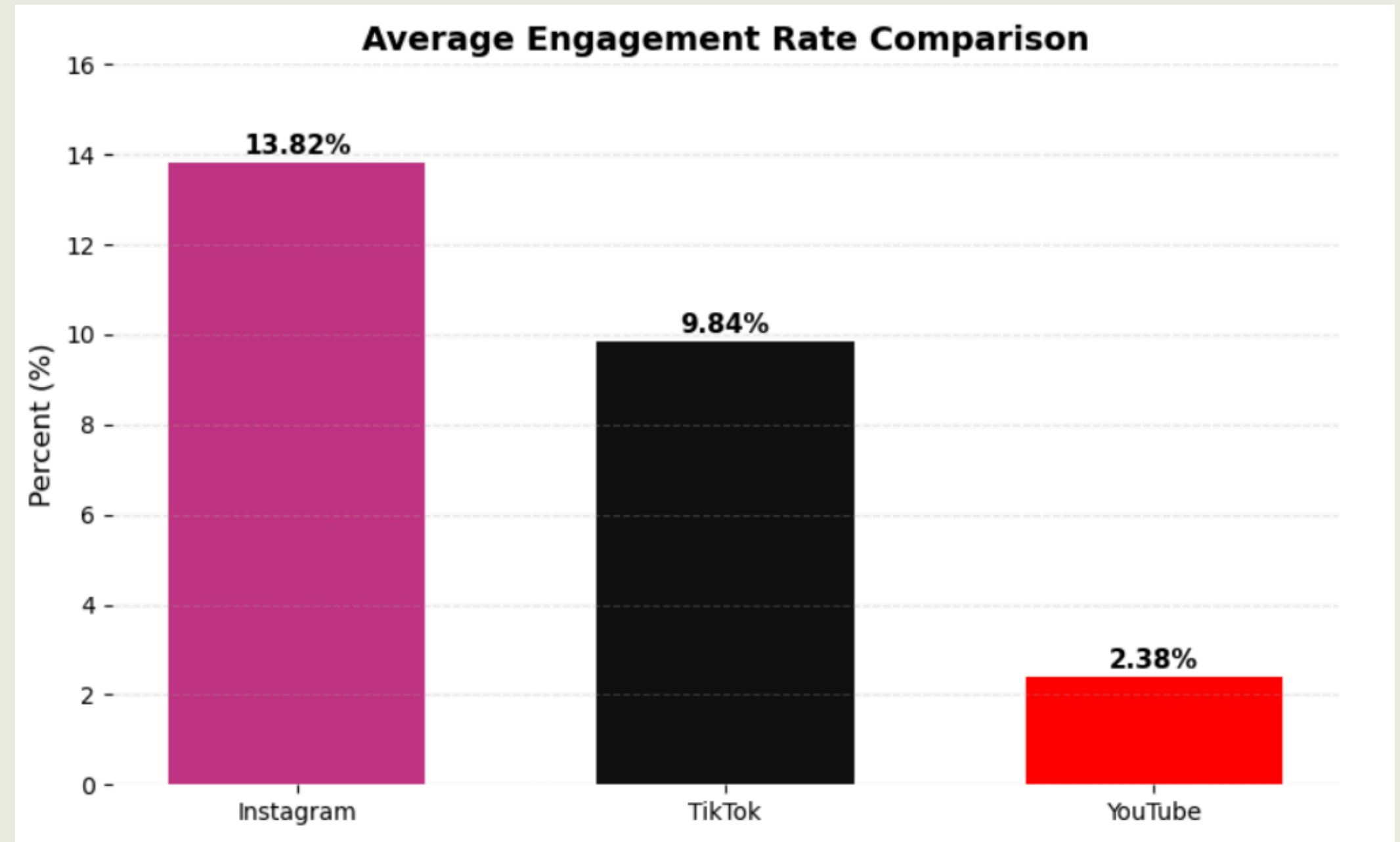
- Instagram achieved the highest engagement rate at 13.82%
- This was 40% higher than TikTok (9.84%)
- It was nearly 6x higher than YouTube (2.38%)

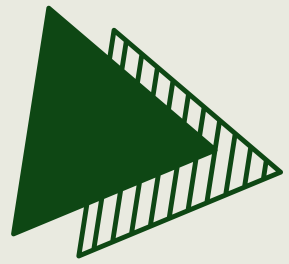
METHODOLOGY

- 300 samples each platform
- Standardized formula
 - **ER = (Likes + Comments) / Exposure**

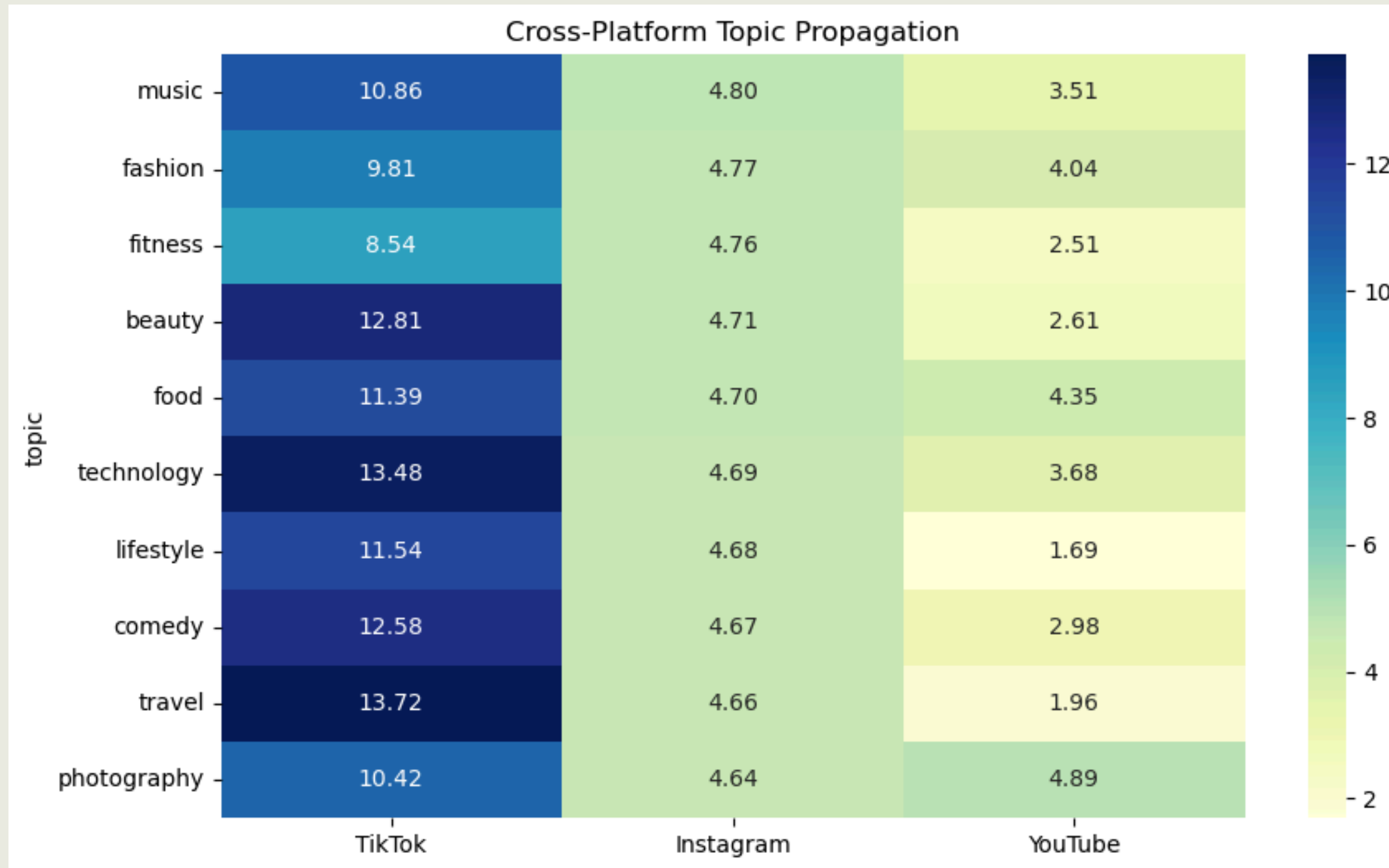
INTERPRETATION

- Instagram may benefit from stronger follower-based interaction
- TikTok performs well through discovery traffic
- YouTube engagement may be more niche-focused



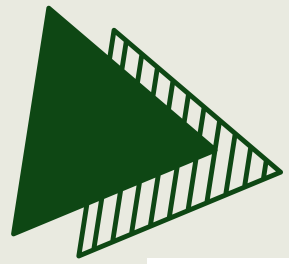


Same Topic, Different Platform Outcome



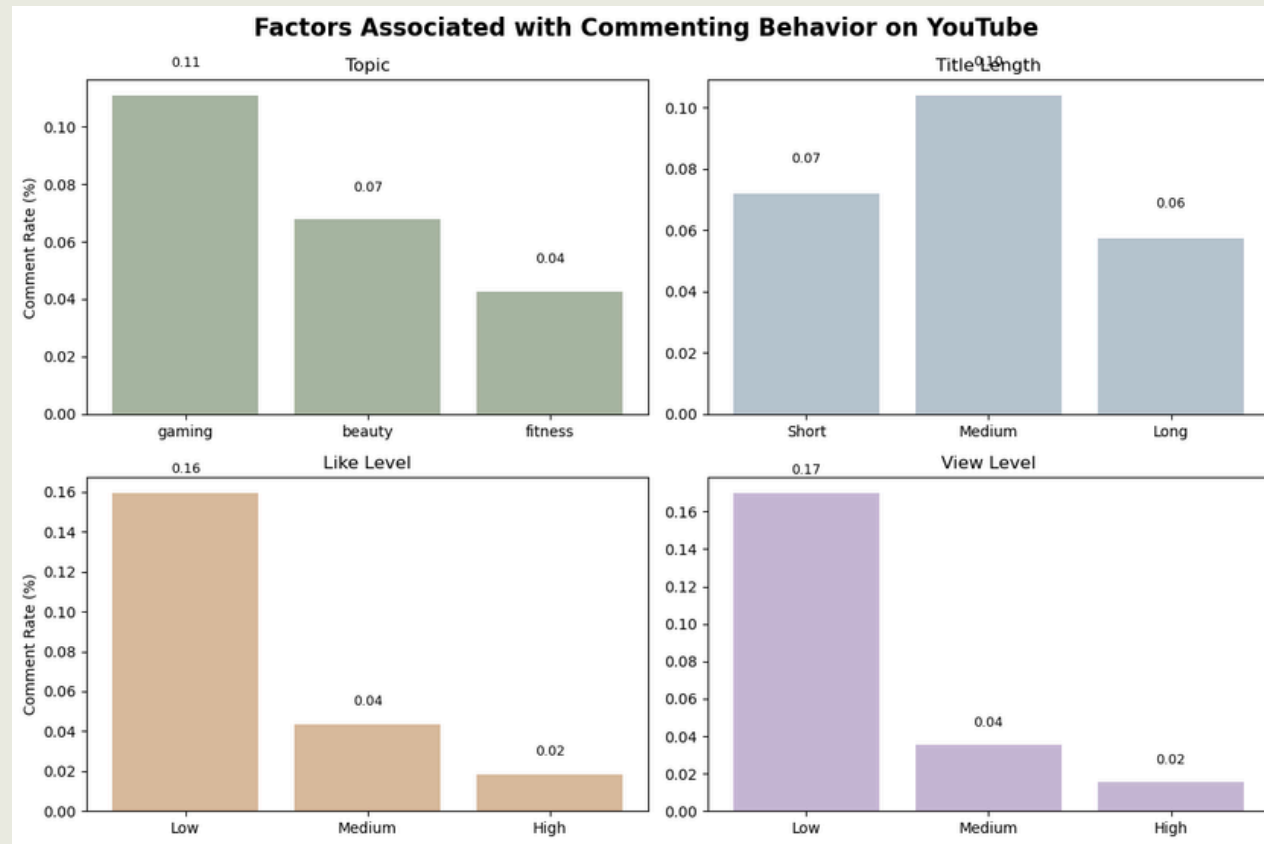
KEY FINDING

- TikTok shows the strongest topic sensitivity
 - Topic engagement ranges from 8.54% to 13.72%
 - Travel and technology perform best
- Instagram remains highly stable
 - All topics stay between 4.64% and 4.80%
 - Topic choice has limited impact
- YouTube favors selective content niches
 - Photography leads at 4.89%
 - Lifestyle is lowest at 1.69%



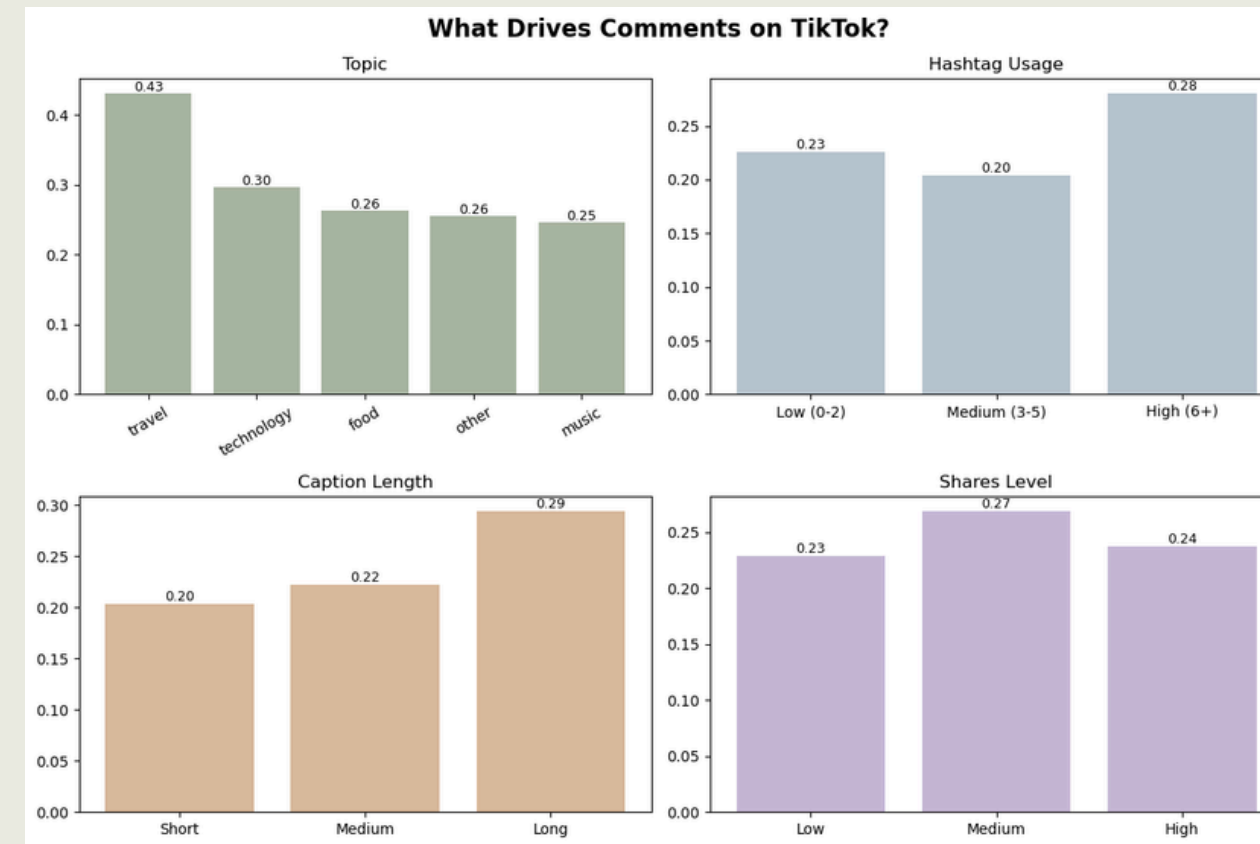
Comment Behavior Comparison

TIKTOK SHOWS THE LARGEST VARIATION

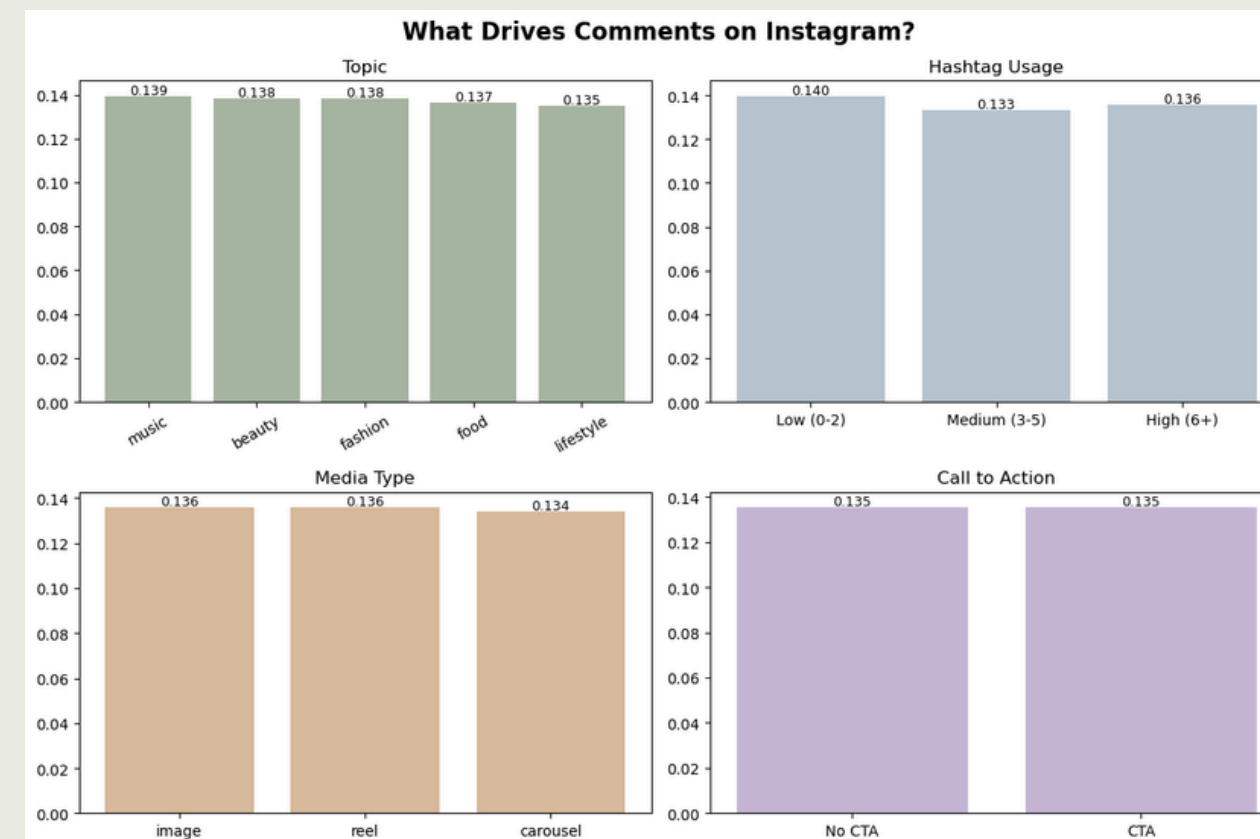


YOUTUBE FAVORS NICHE PARTICIPATION

- Gaming reaches 0.11%
- High-view videos drop to 0.02%

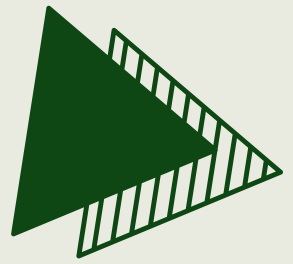


INSTAGRAM REMAINS HIGHLY STABLE



- Comment rates range from 0.20% to 0.43%
- Topic and hashtag effects are strongest

- Most factors stay between 0.133% and 0.140%
- Tactical features have limited impact



Challenges

DATA AVAILABILITY

- Public datasets had different structures across platforms
- Some useful variables were missing or incomplete

METRIC CONSISTENCY

- Engagement metrics were not naturally identical across platforms
- TikTok / YouTube relied on views, while Instagram used reach
- Standardization reduced differences but could not fully remove platform bias

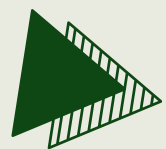
PLATFORM DIFFERENCES

- Each platform has unique algorithms and user behavior patterns
- Higher engagement on one platform does not guarantee the same result elsewhere

TOPIC ALIGNMENT

- Similar topics were manually grouped across datasets
- Some categories may not perfectly match across platforms

Thank you.



Fanshiqi Zhao